

UNIVERSITÄT
LEIPZIG

12 – GGR – M - PG01 - Environmental Change and Natural Risks - A Biogeographical Perspective

3. Species distribution modeling - I

Leipzig, 17.11.2025

Claudia Guimarães-Steinicke
E-Mail: claudia.steinicke@uni-leipzig.de

Schedule

Dates	Content
21.10.2025	Introduction and seminar structure
3.11.2025	Multivariate analysis applied to tropical plant communities - Study case: Atlantic Rain Forest
17.11.2025	**Species distribution modeling (SDM) – Part 1**
8.12.2025	Species distribution modeling (SDM) – Part 2
15.12.2025	Invasive species
8.1.2026	Natural risk : How do fire, windblow, storms influence plant mortality?
20.1.2026	Biodiversity loss and land use change

The goals of this seminar are:

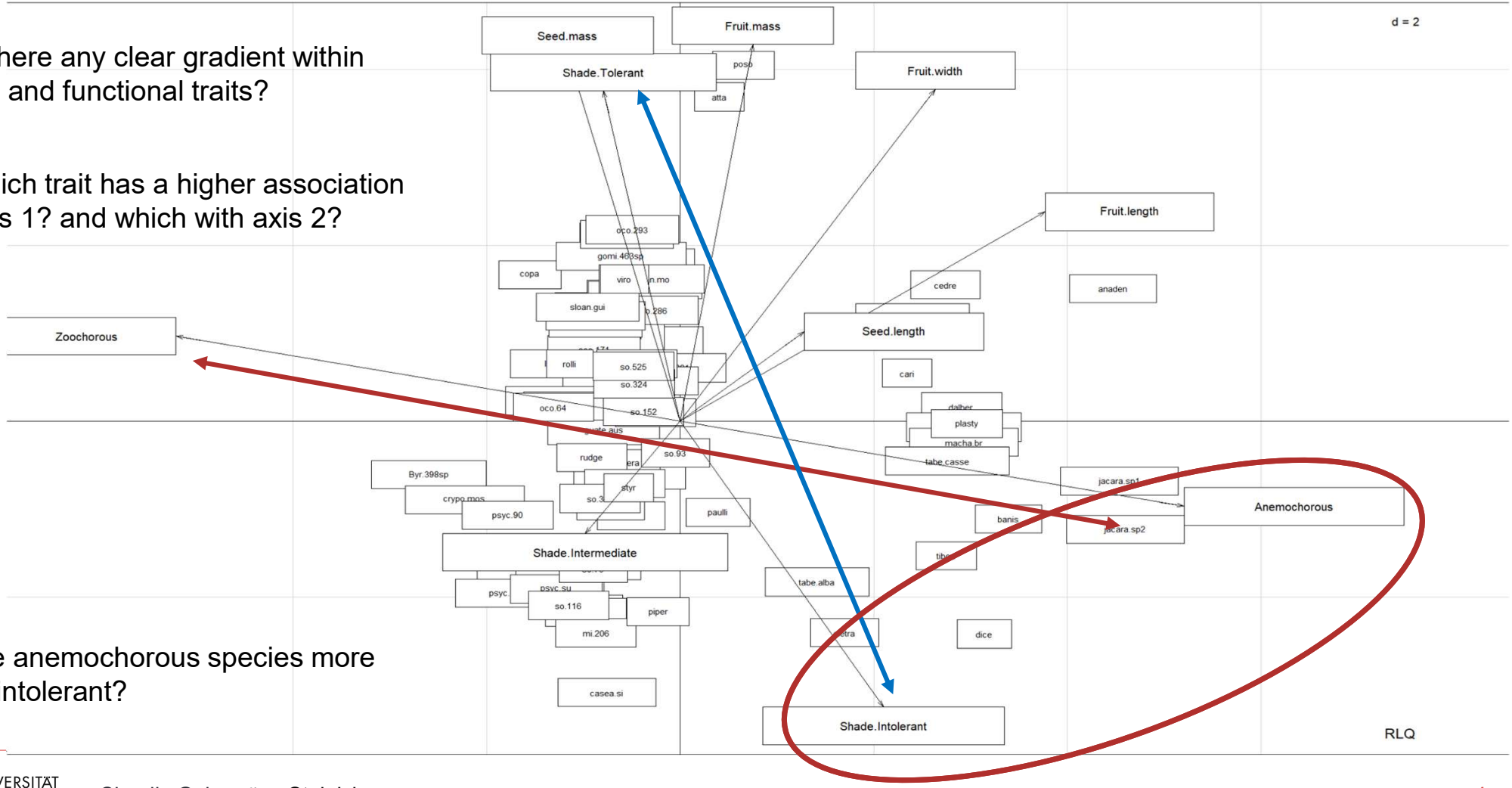
- Recap assignment
- Recap Lecture
- What drives species distributions?
- From niche theory to species distribution
- Types of model´s envelopes
- Basic models principles, and application of the BIOCLIM model

RECAP THE PCA ASSIGNMENT

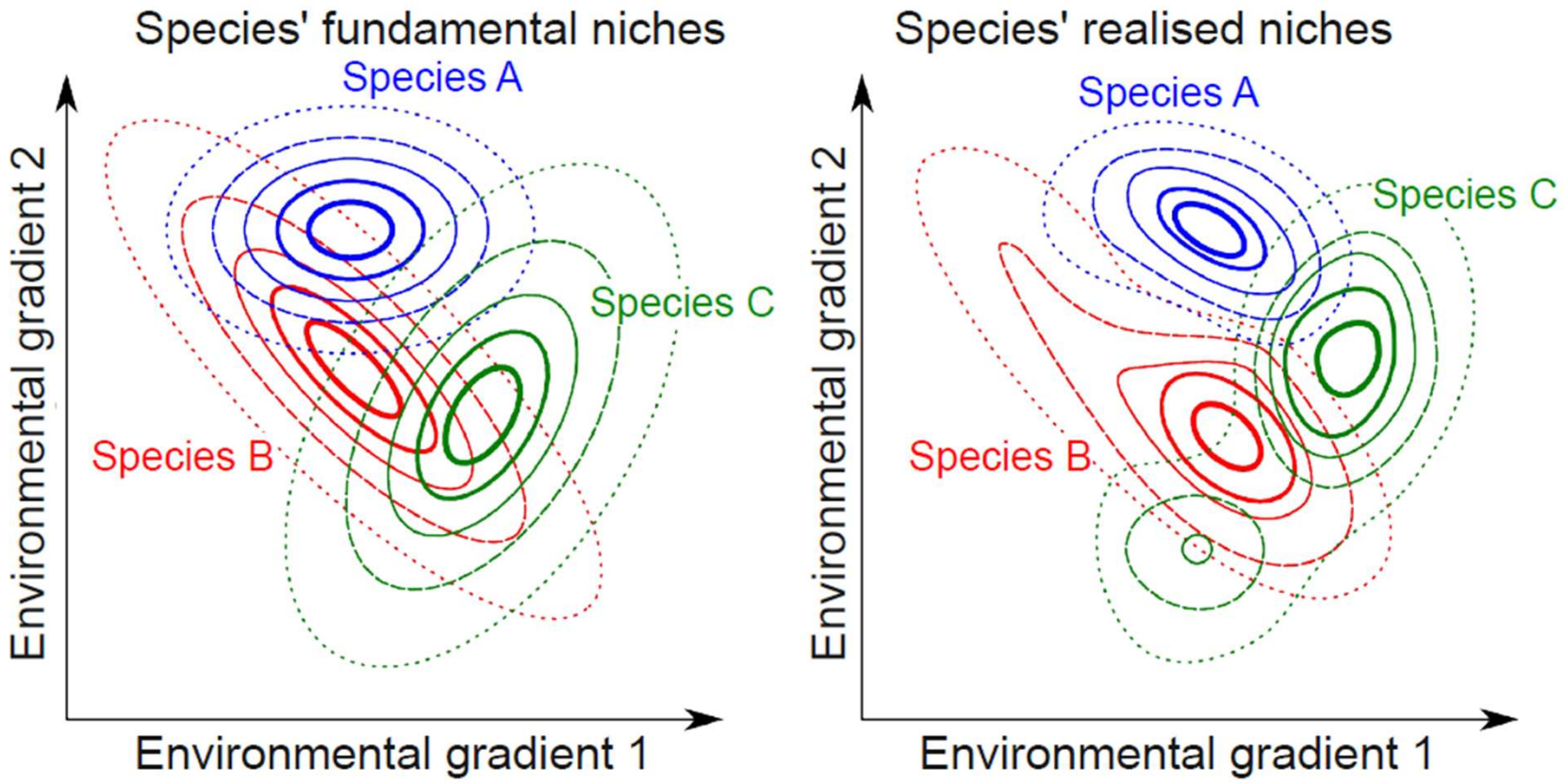
Q1: Is there any clear gradient within species and functional traits?

Q2: Which trait has a higher association with axis 1? and which with axis 2?

Q3: Are anemochorous species more shade-intolerant?



RECAP FROM THE LECTURE





Alexander von Humboldt

Travel to South America

1799 - 1800





Augustin-Pyramus de Candolle

Elementary Theory of Botany

1813

THÉORIE

ÉLÉMENTAIRE

DE LA BOTANIQUE,

OU

EXPOSITION DES PRINCIPES

DE LA CLASSIFICATION NATURELLE

ET DE L'ART

DE DÉCRIRE ET D'ÉTUDIER LES VÉGÉTAUX;

PAR M.r A.-P. DE CANDOLLE,

Professeur d'Histoire Naturelle à l'Académie de Genève, Directeur du Jardin Botanique, Correspondant de l'Institut de France, des Académies Royales des Sciences de Munich, de Turin, du Gard, des Sociétés Phytographique de Gorenki, Helvétique des Sciences Naturelles, Physique de Zurich, Philomatique de Paris, Physiographique de Lund, de Physique et de Chimie d'Arcueil, des Arts, des Sciences Physiques, des Naturalistes et Médico-Chirurgicale de Genève, de la Faculté de Médecine de Paris, de Médecine de Marseille, d'Agriculture de la Seine et de l'Hérault, des Sciences Lettres et Arts de Montpellier et de Rouen, du Lycée d'Histoire Naturelle de New-York, etc., etc.

SECONDE ÉDITION, REVUE ET AUGMENTÉE.

A PARIS,

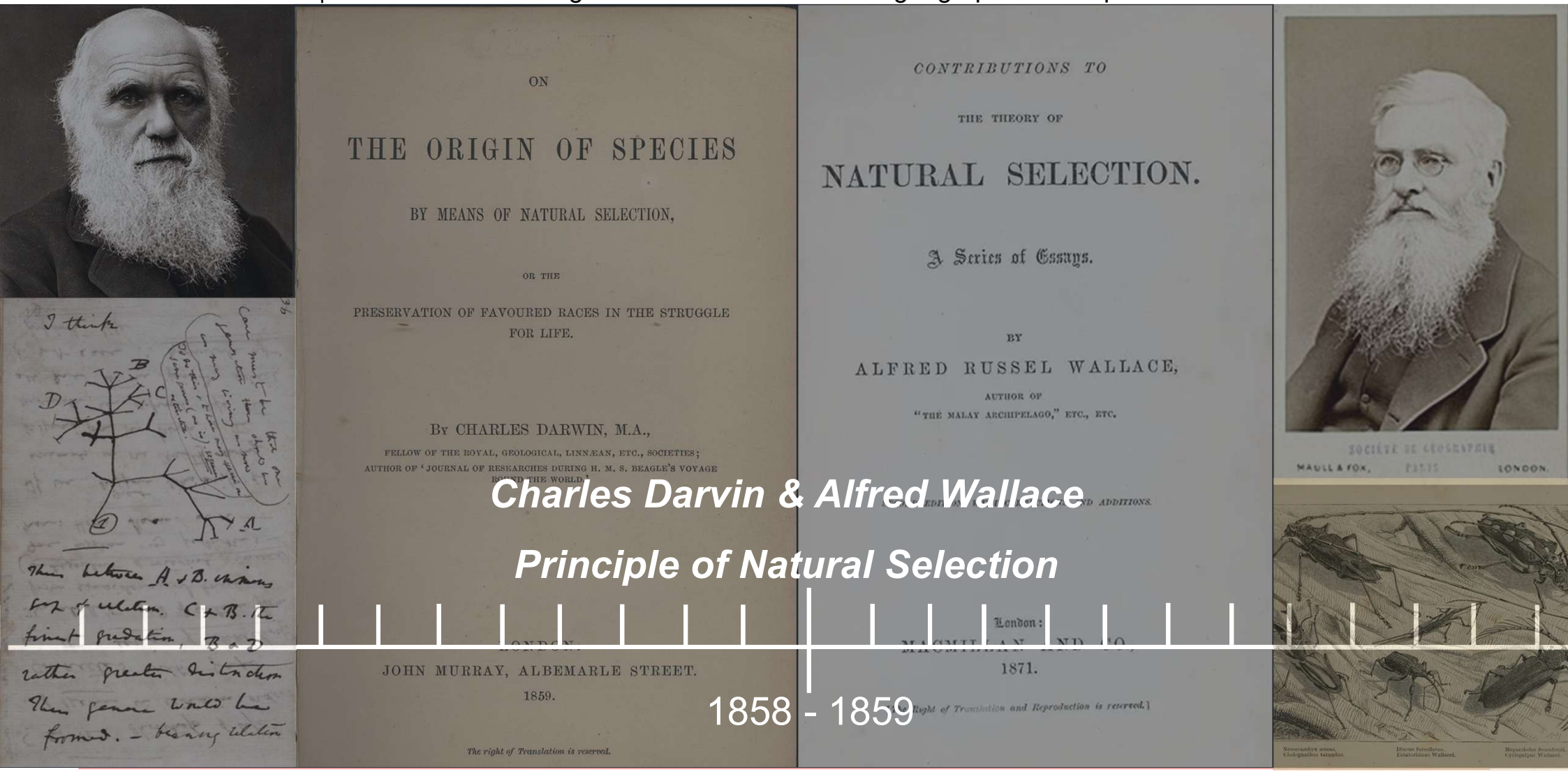
CHEZ DETERVILLE, libraire, rue Hautefeuille, n.° 8.

IMPRIMERIE DE LEBLANC, rue de l'Abbaye, n° 3.

1819.

1938/300

267 F 22

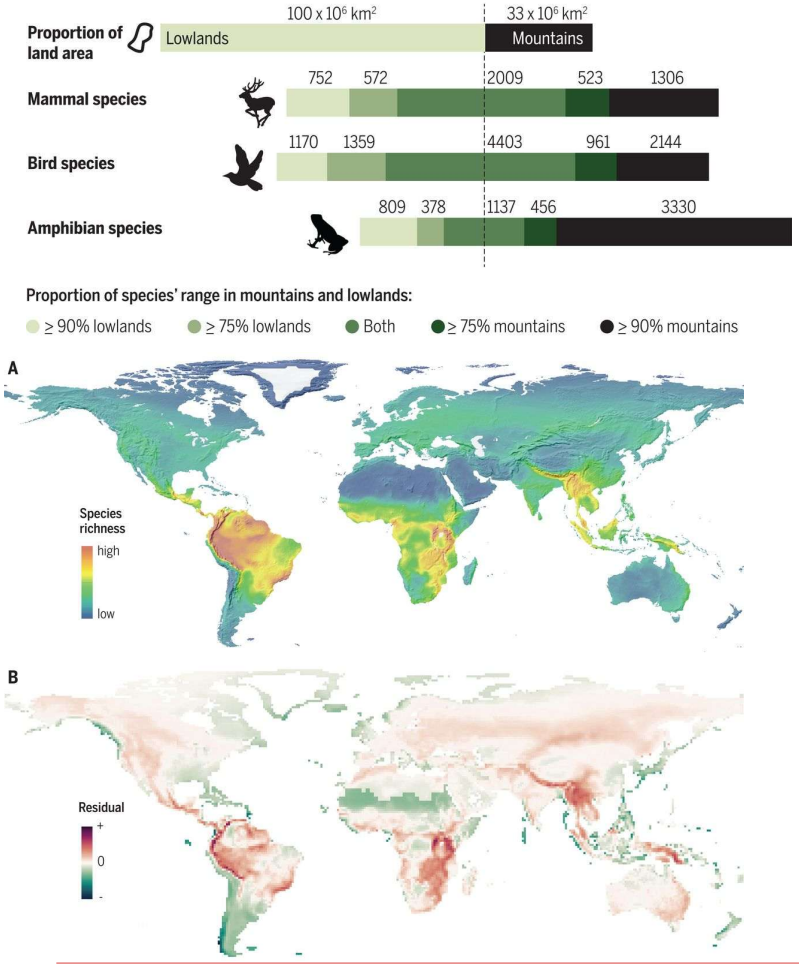


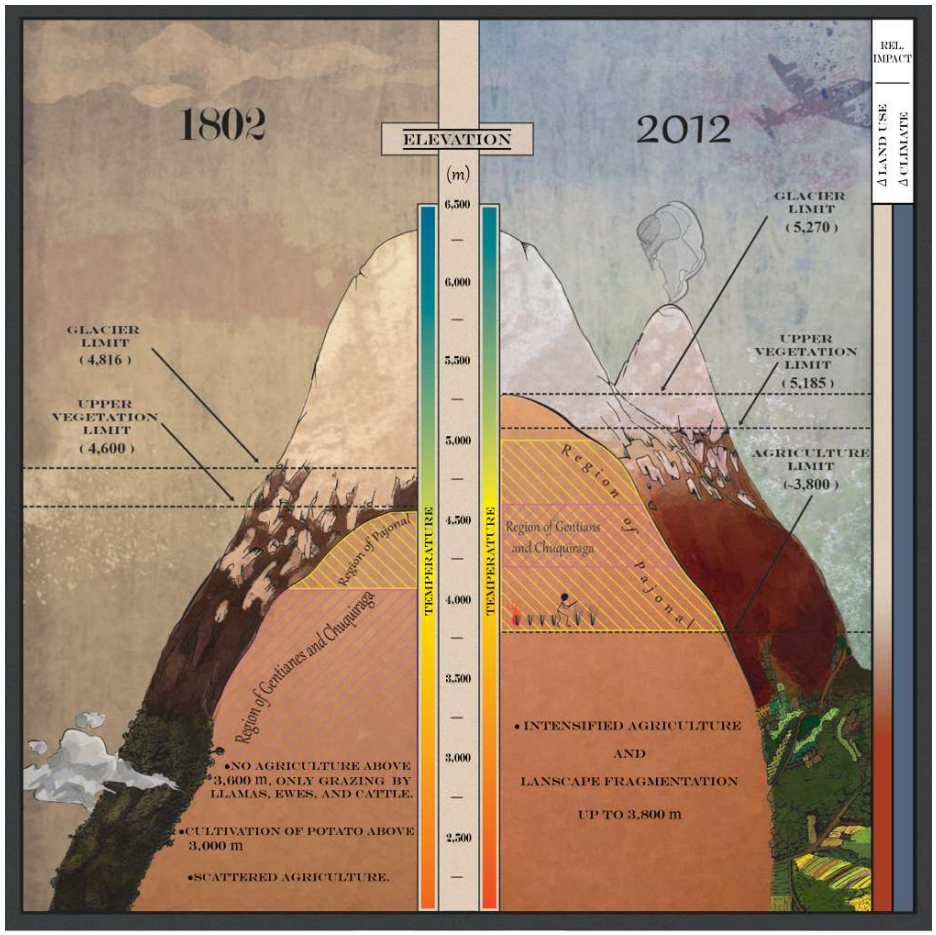
Charles Darvin & Alfred Wallace

Principle of Natural Selection

1858 - 1859

What drives species distributions?

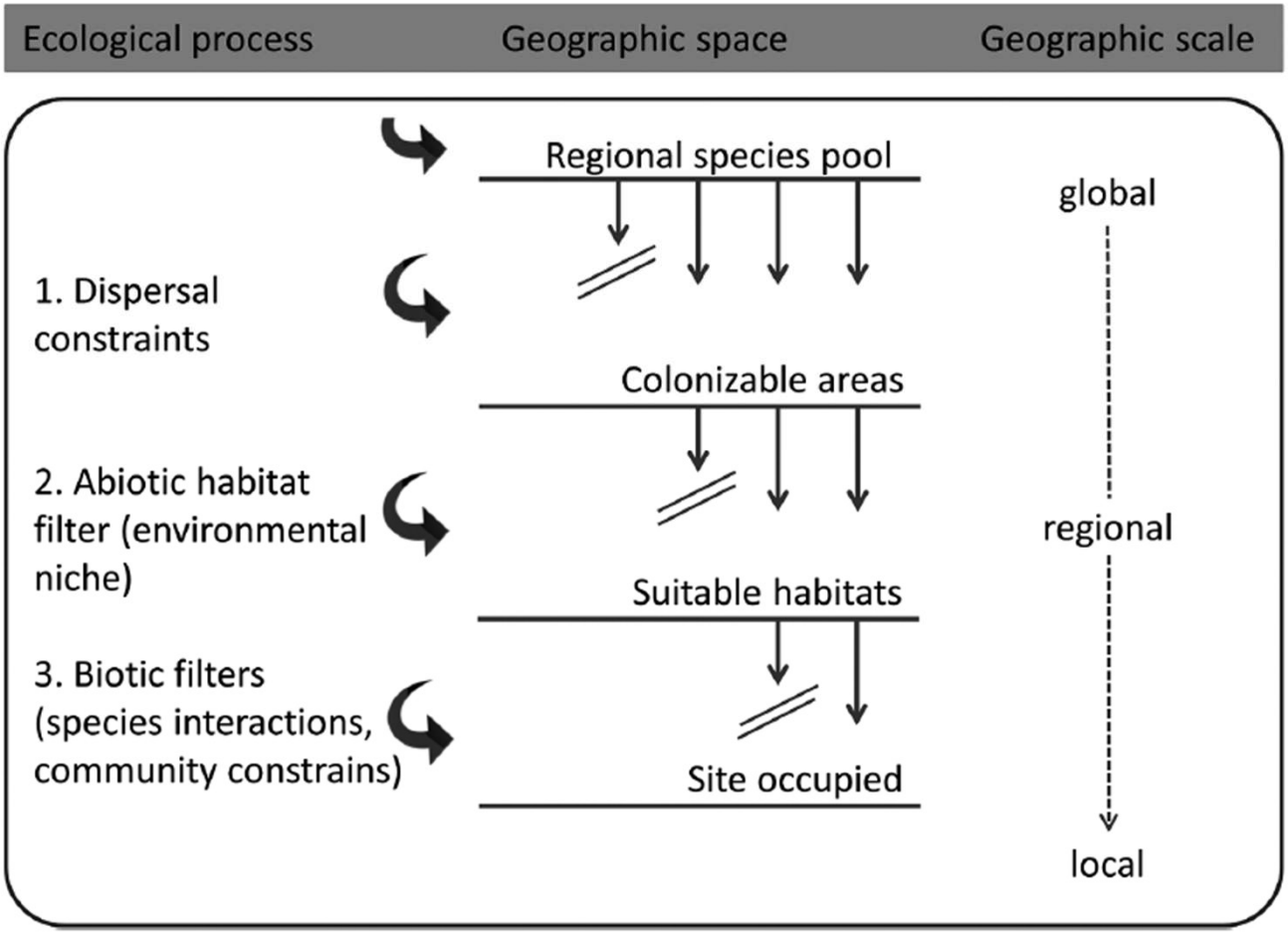




Morueta-Holme et al 2022 (PNAS)

What drives species distributions?

1. Dispersal ability
2. Habitat suitability
3. Biotic interactions



Guisan, Thuiller and Zimmermann 2017

Niche theory

1. **Fundamental niche**: all abiotic conditions where species can survive (pop. Growth) (absence of biotic competition)
2. **Realized niche**: species can survive despite competitors (negative interspecific interactions)
3. **Dispersal ability:** species presence in all suitable habitats



GE Hutchinson

Grinell and Hutchinson 1957, Soberon 2007, Pulliam 1988, Hanski 1998

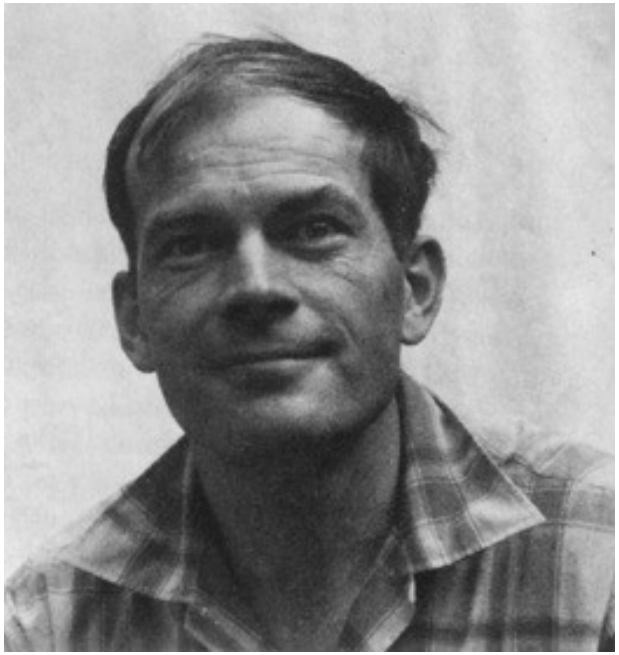
Species range distribution

'Future [biogeographic] theory will concentrate on the boundaries of species ranges as they are encountered on ecologically uniform or continuously varying terrain'

1. Single species range: **habitat specialization**

2. **Why has species not adapted to a wider range of habitats?**

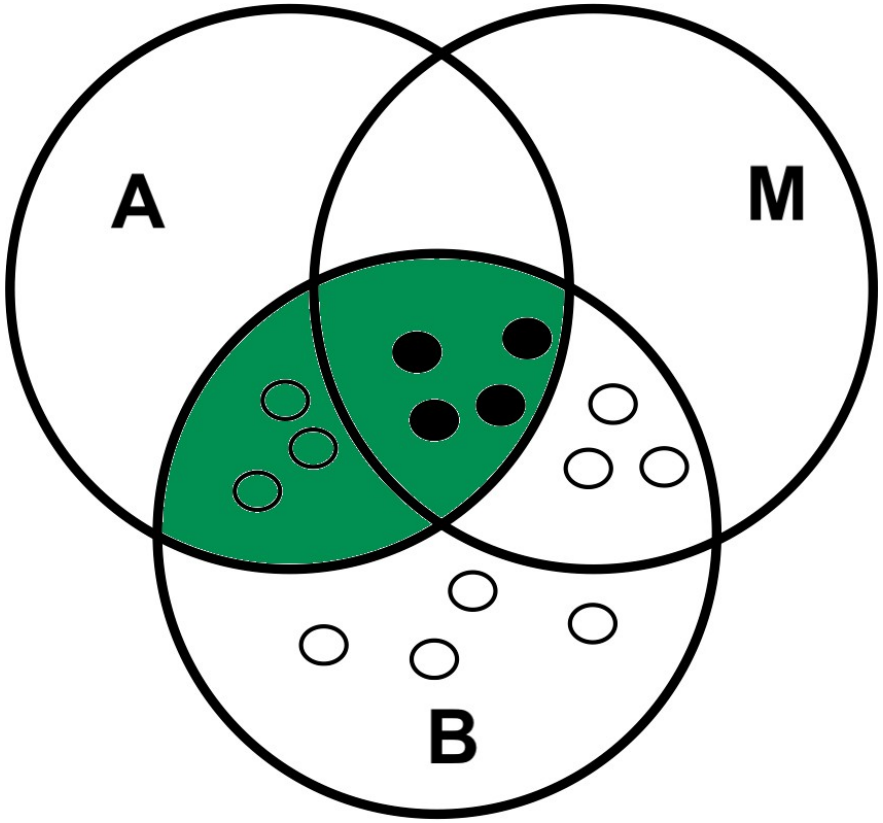
   …It involves adaptive trade-offs between habitats



Robert MacArthur (1972)

1) Why species occurs in some places but not in others?

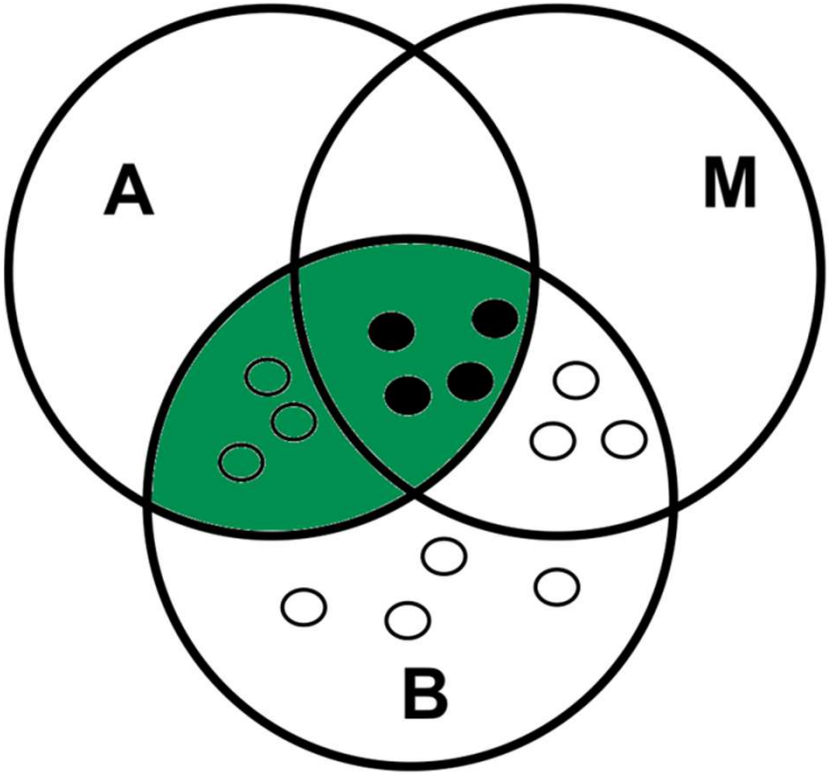
2) Where else we could expect to find the species?



BAM Diagram
Biotic – **A**biotic - **M**ovement

Soberon 2007

Species distribution modeling – Debate on the estimates



1) Near to fundamental niche (Coarse-scale climate data) (presence-only)

2) Realized niche (presence-absence)

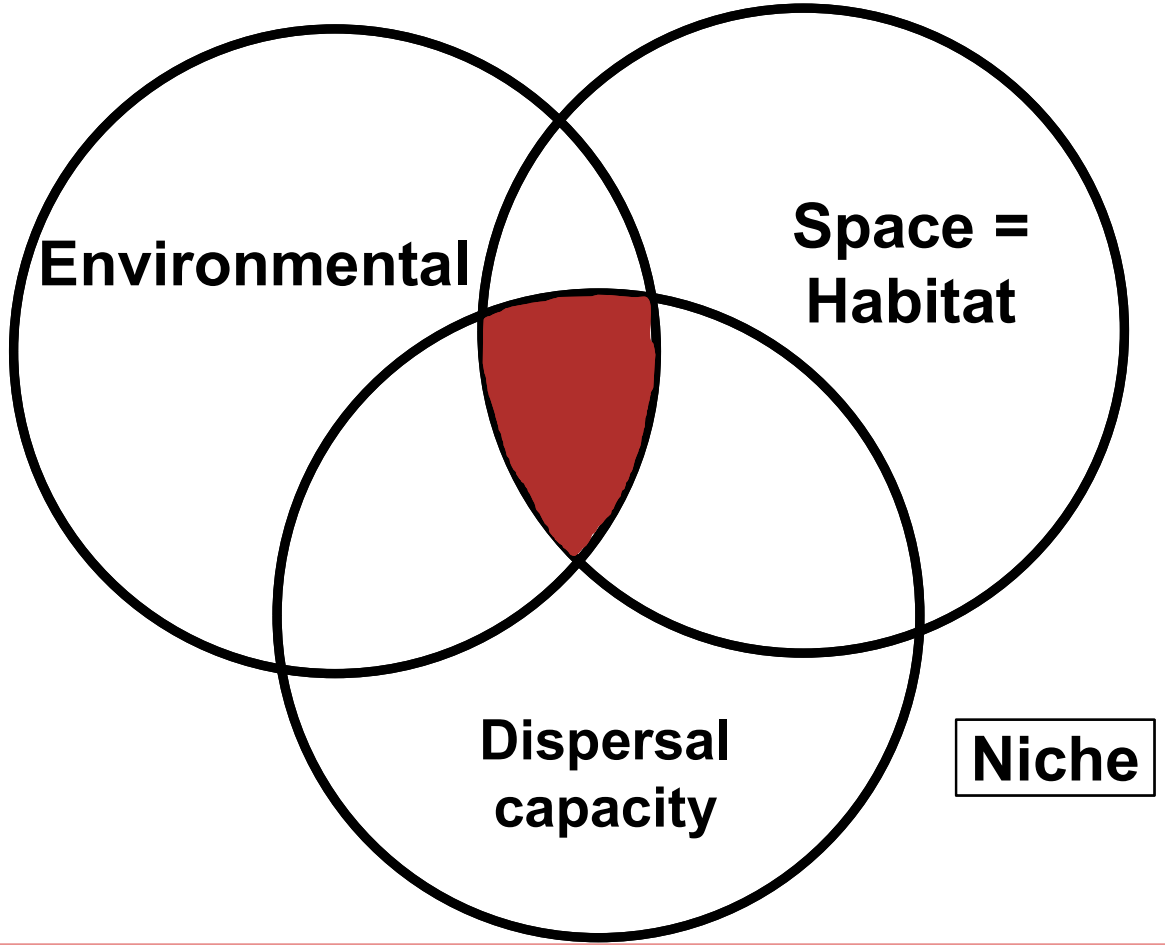
3) Occupied niche (presence-absence)

Debate is based on different envelope models:

“Models use the current geographical distribution of a species to infer its environmental requirements.”

Hijmans and Graham 2006; Zurell and Engler 2019

Causes of species range limits

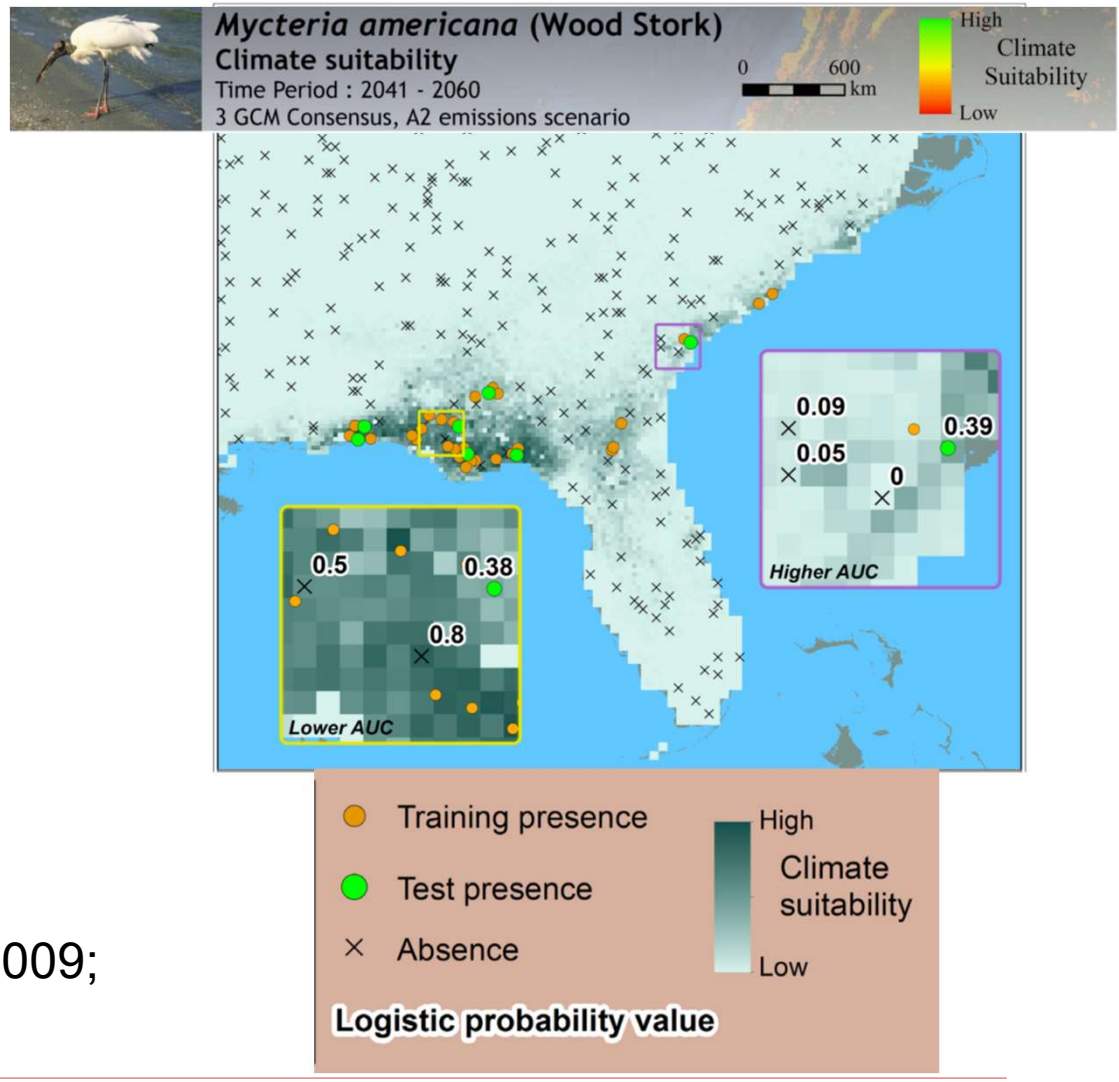


"Species' ranges thus can potentially reflect evolution both in niches and in dispersal."

Brown and Lomolino,1998; Holt 2003

Types of model´s envelope

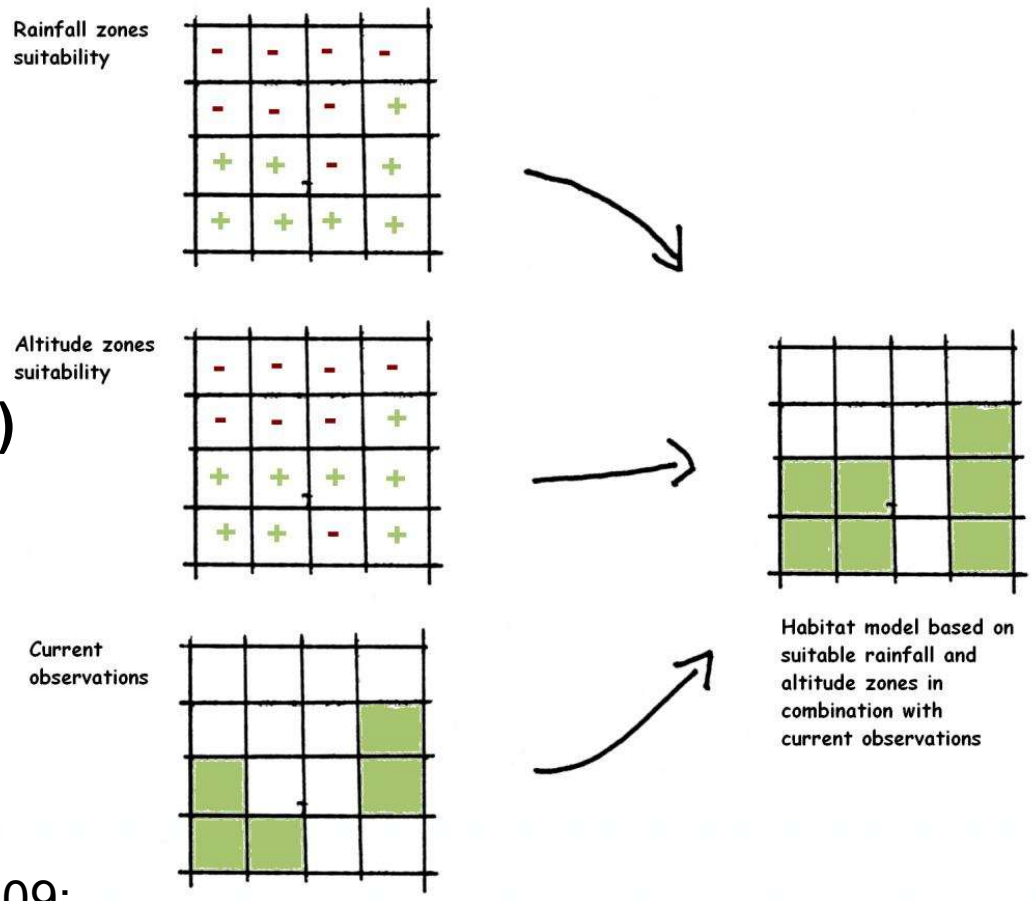
- **Climate envelope (presence-only)**
- Habitat envelope
- Resource selection functions



Watling et al. 2013; Elith and Leathwick 2009; Zurell and Engler 2019

Types of model´s envelope

- Climate envelope
- **Habitat envelope (presence-absence)**
- Resource selection functions

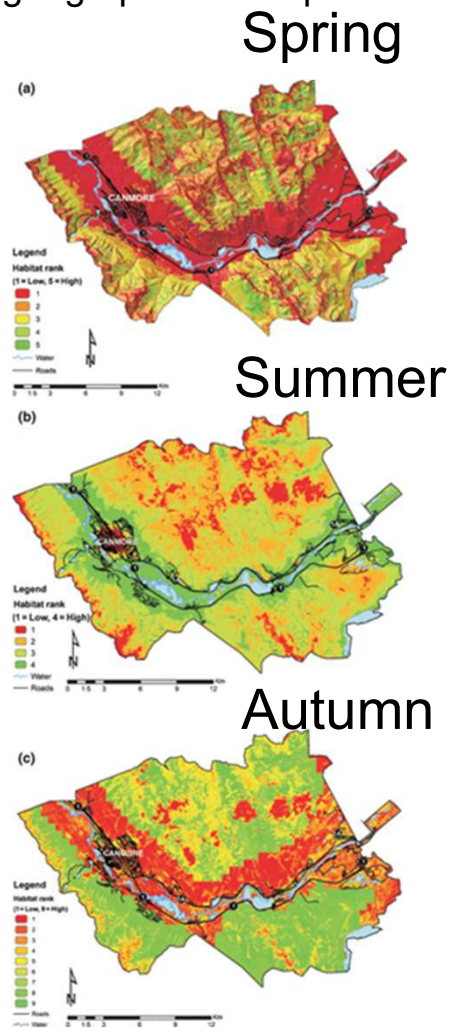


Watling et al. 2013; Elith and Leathwick 2009; Zurell and Engler 2019

Types of model´s envelope

- Climate envelope (Bioclim) (presense-only)
- Habitat envelope (presence-absence)
- **Resource selection functions (presence-absence)**

Location of a species and data on resource availability



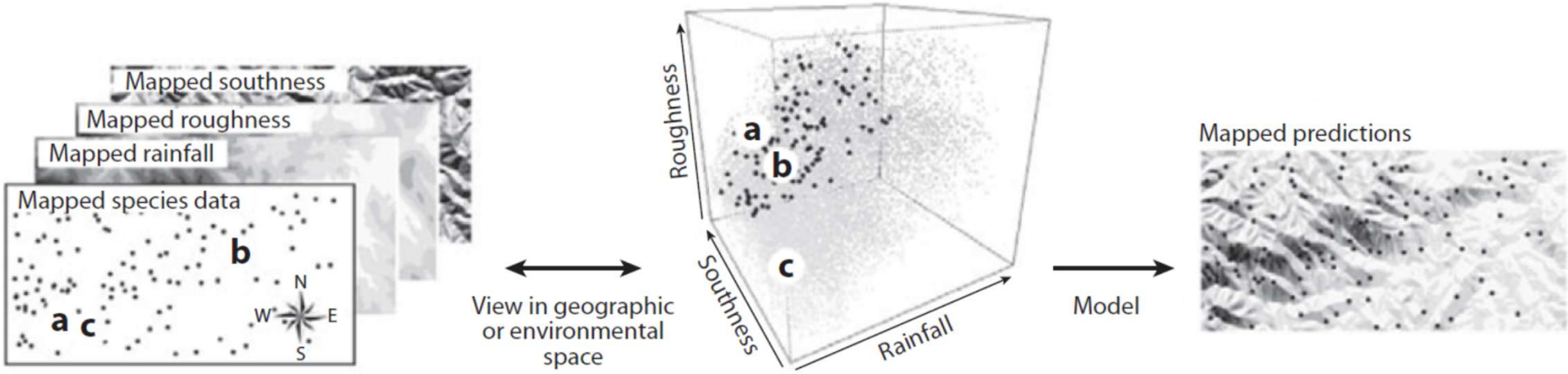


Ursus arctos horribilis

Predicted probability of grizzly bear occurrence in the Canmore, Canada.

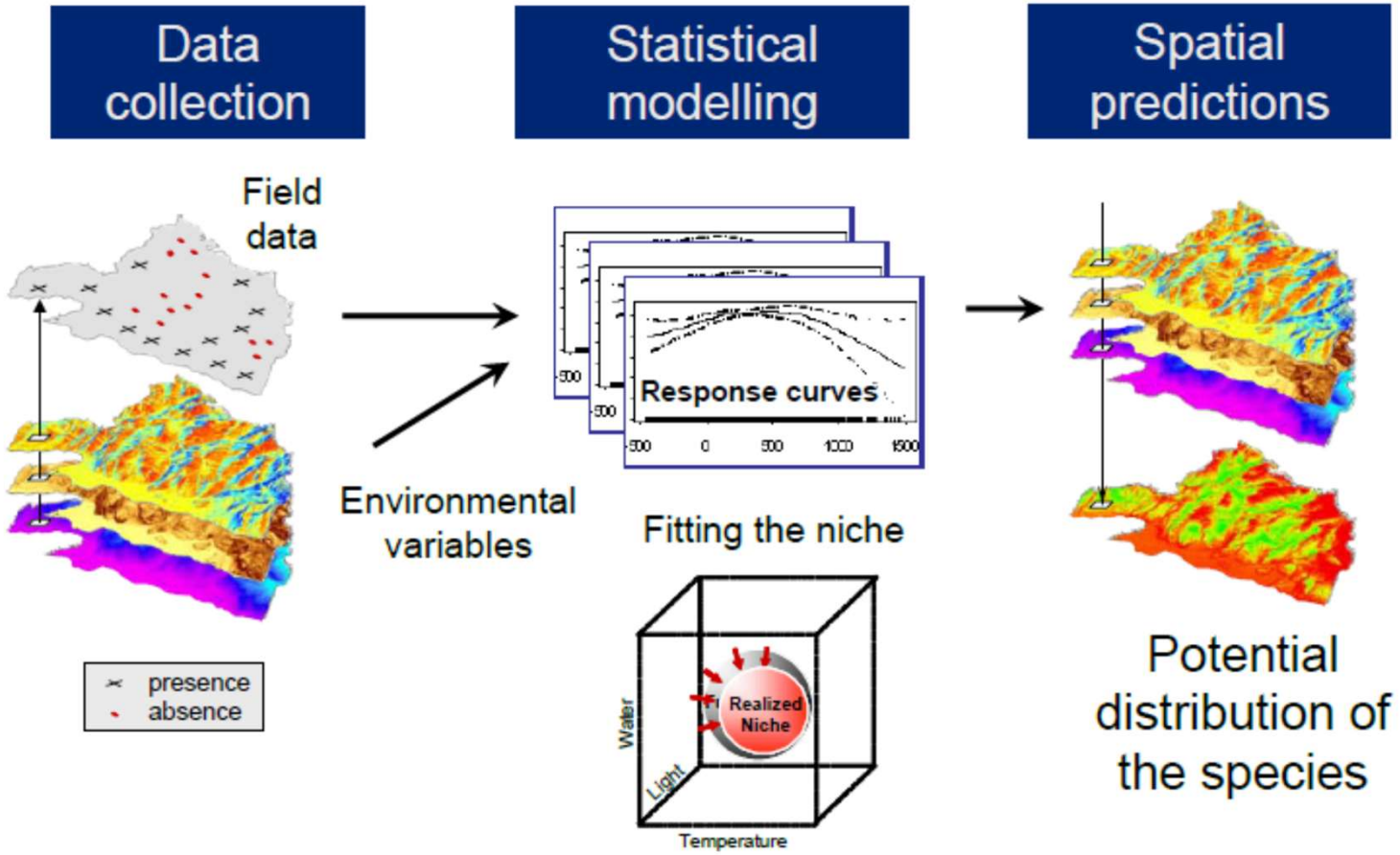
Watling et al. 2013; Elith and Leathwick 2009; Zurell and Engler 2019; Boyce et al. 2002; Cheryl-Lesley et al. 2009

What is a species distribution model?

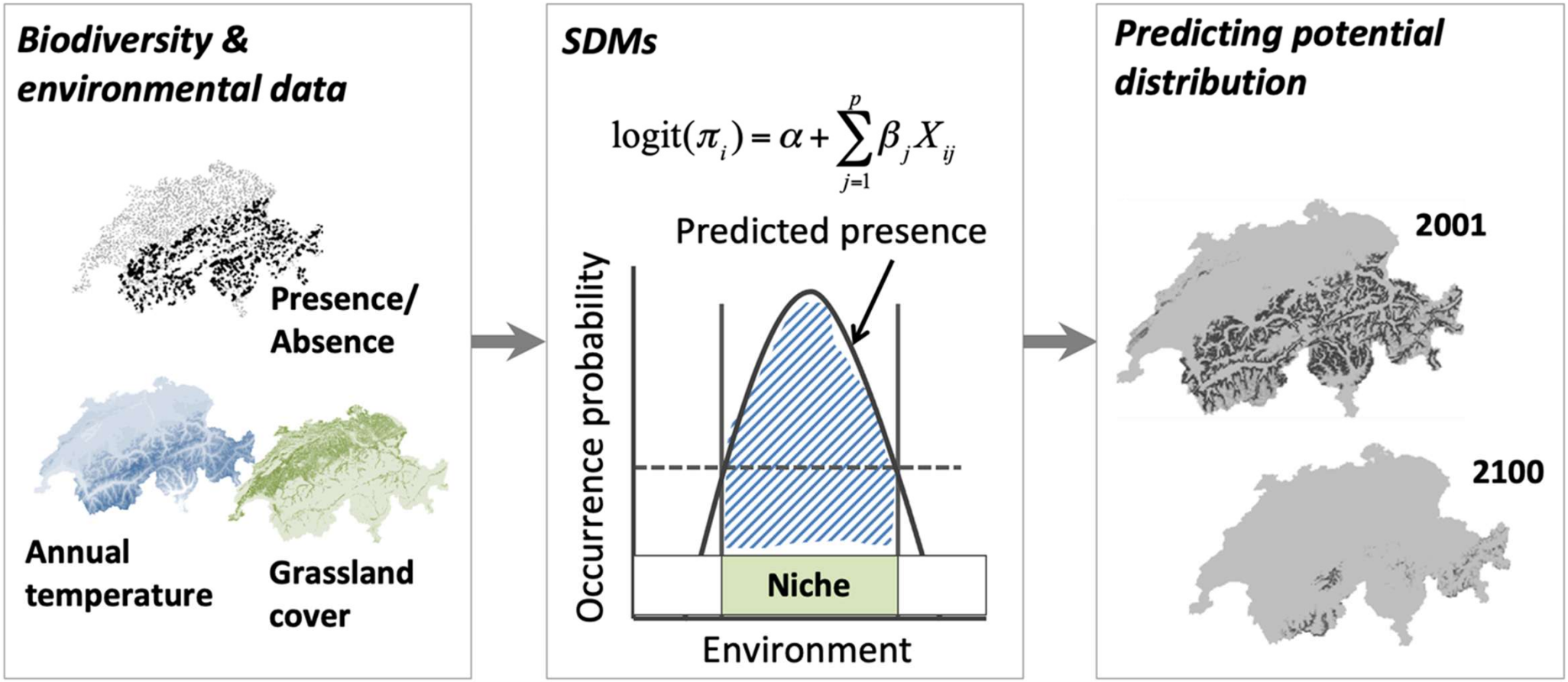


Elith and Leathwick 2009

The basic principle of Species Distribution Model (SDM)



Guisan & Zimmermann 2000 Ecol. Modelling; Zurell and Engler 2019


Biodiversity & environmental data
Presence/ Absence
Annual temperature
Grassland cover
SDMs
$\text{logit}(\pi_i) = \alpha + \sum_{j=1}^{p} \beta_j X_{ij}$
Predicted presence
Occurrence probability
Niche
Environment
Predicting potential distribution
2001
2100

The basic principle of SDM: Steps

Research question - What are the climatic factors that influence the distribution of an endemic tree species in Brazil?







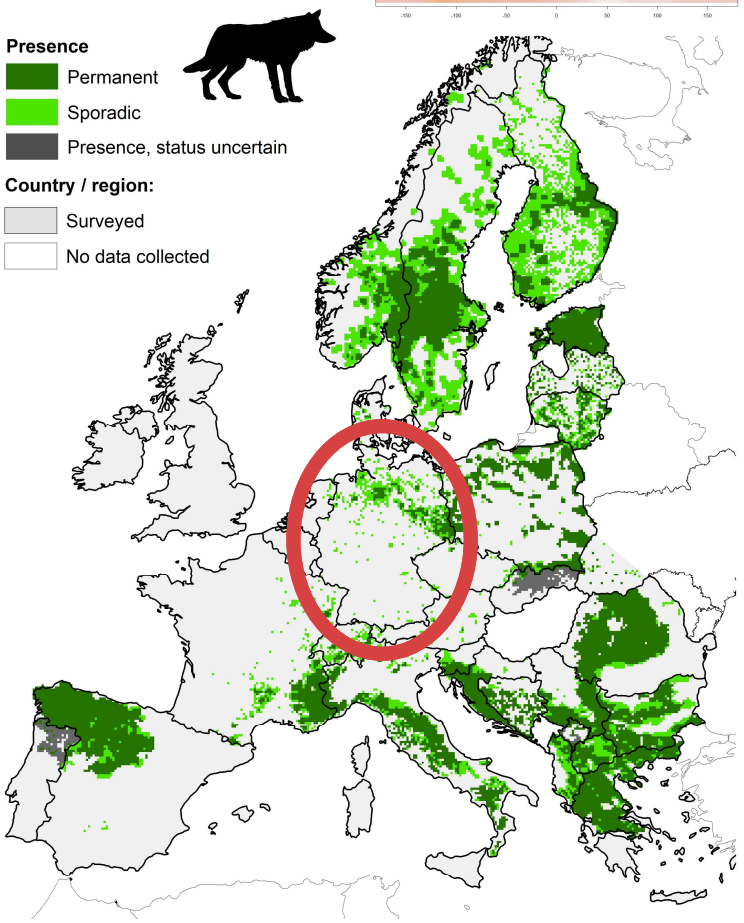


Dalbergia nigra

The basic principle of SDM: Steps

Research question – Which abiotic factor most influences wolf distribution in Germany?





Source: Large Carnivore Initiative for Europe (IUCN)

The basic principle of SDM: Steps

Selection of the model

Presence- absence (background)	Presence-only data
Maxent – Maximum Entropy models	**Bioclim**
CART – Classification-like models	Domain
GLM, GAM – Regression-like models	Habitat
RandomForest - Averaging models	ENFA
ANN, GARP – machine learning models	PCA species

BIOCLIM

"Species have environmental envelopes defined by the environmental conditions at known occurrence locations."

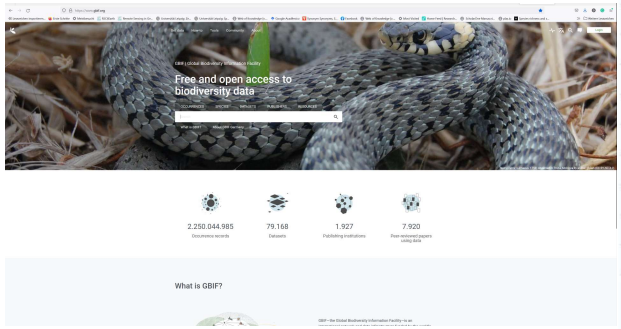
- BIOCLIM describes the multidimensional niche of species based on "**bioclimatic envelopes**"
- It uses percentiles (5 – 95% of environmental variables to define suitability thresholds
- Predictions will show any location falling inside or outside the multidimensional range of the envelope.

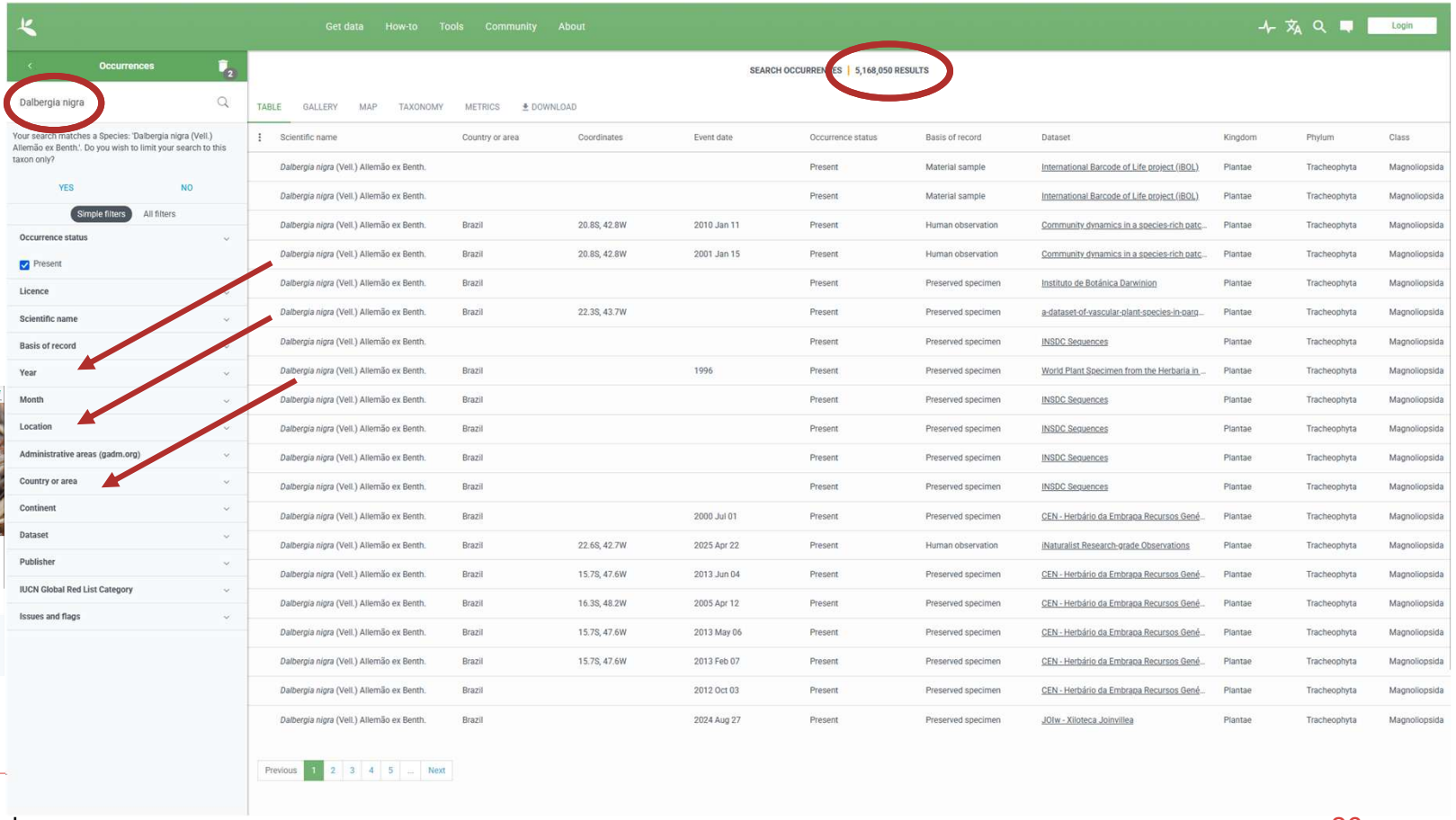
How BIOCLIM works in practice:

1. Data Sampling: GBIF, WWF, etc

https://www.gbif.org

Species occurrence data

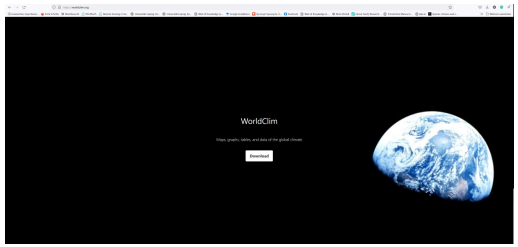




How BIOCLIM works in practice:

2. Extract bioclimatic variables at the location:

Environmental data



https://www.worldclim.org

WorldClim

Global climate and weather data

Welcome to the WorldClim data website.

WorldClim is a database of high spatial resolution global weather and climate data. These data can be used for mapping and spatial modeling. The data are provided for use in research and related activities; and some specialized skill and knowledge is needed to use them (here is some help). More easily available data for the general public will soon be available here.

You can download gridded weather and climate data for historical (near current) and future conditions.

Please let us know if you find a broken link.

© Copyright 2020-2025, worldclim.org.

WorldClim

Historical monthly weather data

From this page you can download historical monthly weather data for 1950-2024. These data are downscaled from CRU-TS-4.09 by the Climatic Research Unit, University of East Anglia, using WorldClim 2.1 for bias correction.

The variables available are average minimum temperature (°C), average maximum temperature (°C) and total precipitation (mm).

The spatial resolution is 2.5 minutes (~21 km^2 at the equator), 5 minutes (~85 km^2) or 10 minutes (~340 km^2). Each download is a "zip" file containing 120 GeoTiff (.tif) files, for each month of the year (January is 1; December is 12), for a 10 year period.

2.5 minutes

years	minimum temperature	maximum temperature	precipitation
1950-1959	tmin_1950-1959	tmax_1950-1959	prec_1950-1959
1960-1969	tmin_1960-1969	tmax_1960-1969	prec_1960-1969
1970-1979	tmin_1970-1979	tmax_1970-1979	prec_1970-1979
1980-1989	tmin_1980-1989	tmax_1980-1989	prec_1980-1989
1990-1999	tmin_1990-1999	tmax_1990-1999	prec_1990-1999
2000-2009	tmin_2000-2009	tmax_2000-2009	prec_2000-2009
2010-2019	tmin_2010-2019	tmax_2010-2019	prec_2010-2019
2020-2024	tmin_2020-2024	tmax_2020-2024	prec_2020-2024

5 minutes

years	minimum temperature	maximum temperature	precipitation
1950-1959	tmin_1950-1959	tmax_1950-1959	prec_1950-1959
1960-1969	tmin_1960-1969	tmax_1960-1969	prec_1960-1969
1970-1979	tmin_1970-1979	tmax_1970-1979	prec_1970-1979
1980-1989	tmin_1980-1989	tmax_1980-1989	prec_1980-1989
1990-1999	tmin_1990-1999	tmax_1990-1999	prec_1990-1999
2000-2009	tmin_2000-2009	tmax_2000-2009	prec_2000-2009
2010-2019	tmin_2010-2019	tmax_2010-2019	prec_2010-2019
2020-2024	tmin_2020-2024	tmax_2020-2024	prec_2020-2024

How BIOCLIM works in practice:

2. Extract bioclimatic variables at the location:

- Biologically meaningful climate summaries
  - i.e., annual mean temperature, precipitation, extreme temperature
- Why are these **derived** climatic variables relevant for ecology than monthly data?
  - Organisms respond to climatic patterns
  - Derived variables reduce the dimensionality and multicollinearity

WorldClim

Bioclimatic variables

Bioclimatic variables are derived from the monthly temperature and rainfall values in order to generate more biologically meaningful variables. These are often used in species distribution modeling and related ecological modeling techniques. The bioclimatic variables represent annual trends (e.g., mean annual temperature, annual precipitation) seasonality (e.g., annual range in temperature and precipitation) and extreme or limiting environmental factors (e.g., temperature of the coldest and warmest month, and precipitation of the wet and dry quarters). A quarter is a period of three months (1/4 of the year).

They are coded as follows:

BIO1 = Annual Mean Temperature
BIO2 = Mean Diurnal Range (Mean of monthly (max temp - min temp))
BIO3 = Isothermality (BIO2/BIO7) (×100)
BIO4 = Temperature Seasonality (standard deviation ×100)
BIO5 = Max Temperature of Warmest Month
BIO6 = Min Temperature of Coldest Month
BIO7 = Temperature Annual Range (BIO5-BIO6)
BIO8 = Mean Temperature of Wettest Quarter
BIO9 = Mean Temperature of Driest Quarter
BIO10 = Mean Temperature of Warmest Quarter
BIO11 = Mean Temperature of Coldest Quarter
BIO12 = Annual Precipitation
BIO13 = Precipitation of Wettest Month
BIO14 = Precipitation of Driest Month
BIO15 = Precipitation Seasonality (Coefficient of Variation)
BIO16 = Precipitation of Wettest Quarter
BIO17 = Precipitation of Driest Quarter
BIO18 = Precipitation of Warmest Quarter
BIO19 = Precipitation of Coldest Quarter

This scheme follows that of ANUCLIM, except that for temperature seasonality the standard deviation was used because a coefficient of variation does not make sense with temperatures between -1 and 1).

To create these values yourself, you can use the 'biovars' function in the *R* package dismo.

How BIOCLIM works in practice:

3. Compute percentile-based limits for each variable

Tolerance range for a species: Lower percentile (5th) and upper percentile (95%)

But why percentiles?

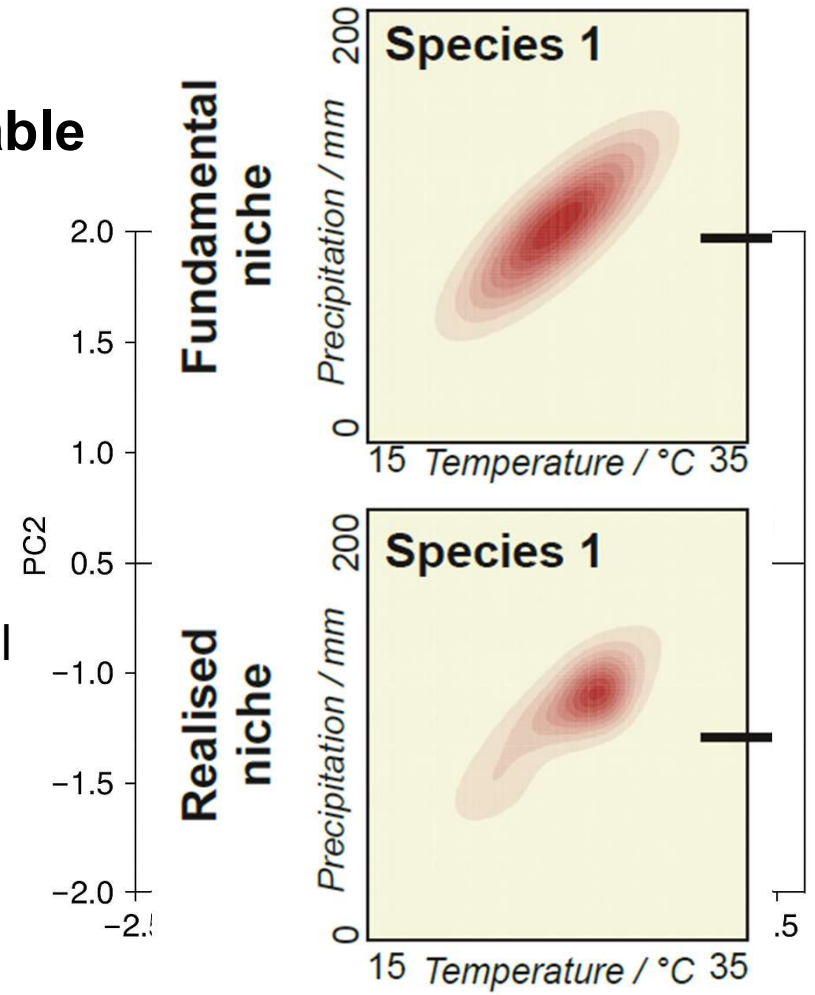
- Robust to outliers, simplicity, and simple interpretation

Ecological meaning:

- Each variable´s percentile range describes physiological tolerance, ecological limits

Multidimensional climate envelope

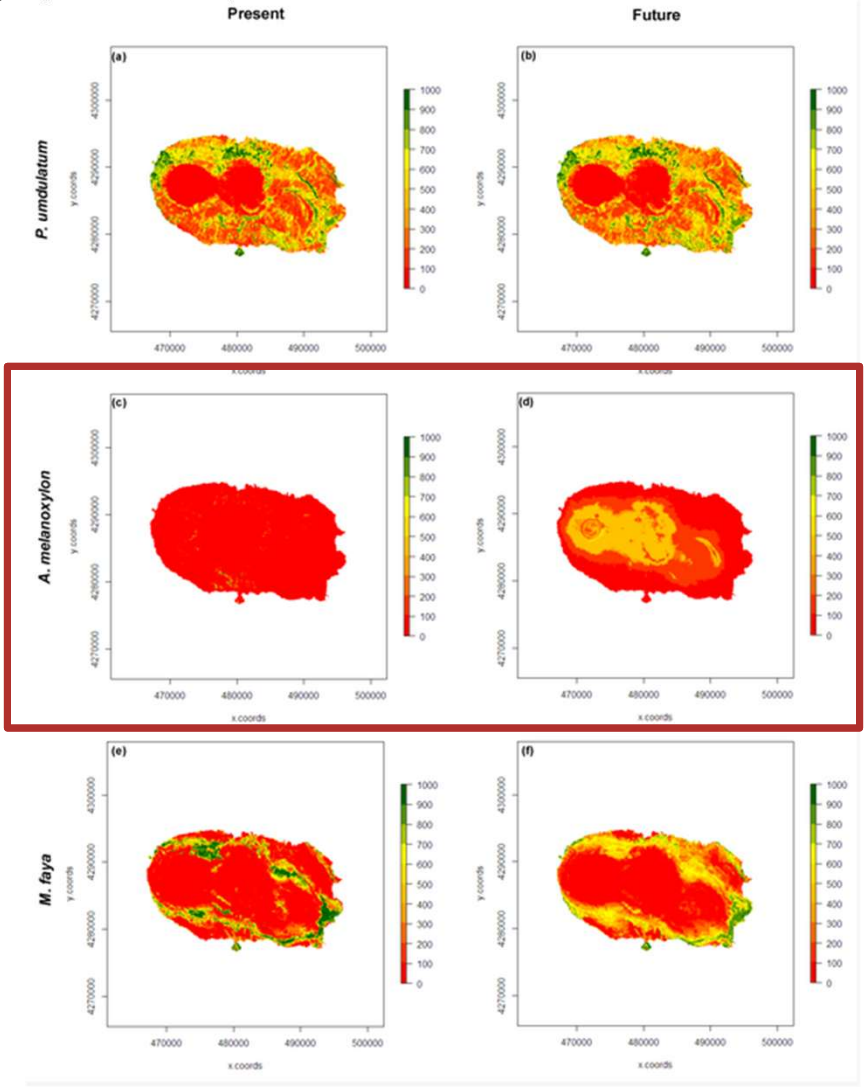
- For BIO1: lower1–upper1
  For BIO12: lower2–upper2
  For BIO5: lower3–upper3



How BIOCLIM works in practice:

4. Limitations

- Rectangular niche assumptions
- Ignores variable interactions
- Sampling bias (based on occurrence)
- Scale & resolution issues
- Poor predictions: climate only
- Extrapolation risks (future scenarios)



Silva et al. 2019

SUMMARY

- What drives species distributions?
  - Dispersal, habitat suitability, and biotic interactions
- Niche theory – Model the species distributions
  - Fundamental and realized niches
- Bioclimatic envelope (first studies)
- BIOCLIM models
  - Bioclimatic envelope for a species in the range (or percentile range)



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R EXERCISES: 3-SDM_1.RMD